

Suppose $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is linear. Then either T has 2 (perhaps equal) real e-values or T has complex conjugate e-values.

The proof of the following proposition uses just linear algebra.

Proposition 1. *Suppose $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is linear with e-values $\lambda e^{\pm i\theta}$ and $\lambda > 0$. Then $\lambda^{-1}T$ leaves some ellipse invariant.*

Proof. Using the same matrix as in the real case, consider T as a (complex) linear map $T : \mathbb{C}^2 \rightarrow \mathbb{C}^2$. Diagonalising, there exists a linear $S : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ such that

$$(1) \quad S(\lambda^{-1}T)S^{-1} = Q = \begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix}.$$

Q fixes the unit sphere

$$\begin{aligned} \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\} &= \{(x_1, y_1, x_2, y_2) : x_1^2 + y_1^2 + x_2^2 + y_2^2 = 1\} \\ &=: S^3 \subset \mathbb{R}^4. \end{aligned}$$

Imbed $\mathbb{R}^2 \subset \mathbb{C}^2 \cong \mathbb{R}^4$ via $(x_1, x_2) \leftrightarrow (x_1, x_2) \leftrightarrow (x_1, 0, x_2, 0)$. Let $V = S(\mathbb{R}^2) \subset \mathbb{R}^4$. Note V is a real 2-dimensional subspace and S is injective onto its image.

Let

$$C = V \cap S^3 \subset V \subset \mathbb{R}^4, \quad E = S^{-1}(C) \subset \mathbb{R}^2.$$

Then C is a unit circle and E is an ellipse.

Then

$$Q(C) = C$$

since $Q(C)$ is a circle from the third term in (1), and $Q(C) \subset V$ from the first term in (1). But there is only one unit circle in V .

Applying S^{-1} to the left of the first two terms in (1), and applying this to C ,

$$\lambda^{-1}TS^{-1}(C) = S^{-1}Q(C) = S^{-1}(C).$$

That is,

$$\lambda^{-1}T(E) = E. \quad \square$$